Course: Linear Algebra and Optimization **Course Code: 23MA4BSLAO**

Unit 5: Matrix Decomposition and their Applications

Similar Matrices: Two $n \times n$ matrices A and B are similar if there is an invertible matrix P such that $B = P^{-1}AP$.

Diagonalization:

A square matrix is diagonalizable if it is similar to a diagonal matrix.

Therefore, the square matrix A is diagonalizable if $A = PDP^{-1}$.
Consider $A = PDP^{-1} \Longrightarrow AP = PD$ and $P = [p_1\ p_2\ p_3 \dots p_n]$
$\Longrightarrow [Ap_1\ \ Ap_2\ \ Ap_3 \dots Ap_n] = [d_1p_1\ d_2p_2\ d_3p_3 \dots d_np_n]$
Hence $p_1, p_2, p_3, \dots, p_n$ are the eigenvectors of A and $d_1, d_2, d_3, \dots, d_n$ are the eigenvalues of A.
This implies that a square matrix is diagonalizable if and only if the matrix is non-defective i.e. (A has n linearly independent eigenvectors)

Sufficient condition for diagonalization: If an $n \times n$ matrix A has n distinct eigenvalues, then the corresponding eigenvectors are linearly independent and A is diagonalizable.

Note:

- Similar matrices have the same eigenvalues.

Working Rule to diagonalize a given square matrix A

1. Determine all the eigenvalues of A and find the corresponding linearly independent eigenvectors $p_1, p_2, p_3, \dots, p_n$.
2. Construct the modal matrix $P = [p_1\ p_2\ p_3 \dots p_n]$ and compute P^{-1}.
3. Compute diagonal matrix $D = P^{-1}AP = diag(\lambda_1, \lambda_2, \dots, \lambda_n)$ where each eigen value satisfies $Ap_i = \lambda_i p_i$.

This diagonalization can be applied to compute integral powers of the matrix A: i.e., $A^n = PD^nP^{-1}$.

I. Find the modal matrix P which diagonalizes the following matrices.

1. $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ also find A^4.
2. $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$.
3. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$.
4. $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$ also find A^6.
5. $A = \begin{bmatrix} 0 & 0 & -2 \\ 0 & -2 & 0 \\ -2 & 0 & 3 \end{bmatrix}$.
6. $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ also find A^4.

Orthogonal Matrix: A square matrix P is called an orthogonal matrix iff $P^{-1} = P^T$.

Theorem: A real square matrix is orthogonally diagonalizable if and only if it is symmetric.

Spectral Theorem: Any $n \times n$ symmetric matrix has the following properties:

1. A has n real eigenvalues, counting the algebraic multiplicities.
2. The dimension of eigenspace of each eigenvalue λ equals the algebraic multiplicities.
3. Eigenvectors corresponding to distinct eigenvalues are orthogonal.
4. A is orthogonally diagonalizable.

Working Rule: Let A be an $n \times n$ symmetric matrix.

1. Find all eigenvalues of A and determine their algebraic multiplicities.
2. Find the corresponding eigenvectors: for distinct eigenvalues take unit eigenvectors, and for repeated eigenvalues obtain a set of linearly independent eigenvectors and orthonormalize them using the Gram–Schmidt process.
3. Form the orthogonal matrix P, using the **orthonormal** eigenvectors as columns.
4. Then $P^{-1}AP = P^{T}AP = D$, where D is a diagonal matrix containing the eigenvalues of A.

Note: An $n \times n$ matrix P is orthogonal if and only if its column vectors form an orthogonal set.

II. Orthogonally diagonalize the following matrices.

1. $A=\begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$.
2. $A=\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.
3. $A=\begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$.
4. $A=\begin{bmatrix} 2 & 2 & 4 \\ 2 & 5 & 8 \\ 4 & 8 & 17 \end{bmatrix}$.

QUADRATIC FORMS:

A function $Q: R^{n} \to R$ is called a quadratic form if $Q(X) = X^{T}AX = \sum_{i,j=1}^{n} A_{ij}X_iX_j$ where A is a symmetric matrix.

Canonical Form:

Let A be the matrix of the quadratic form Q and let λ_1, λ_2 and λ_3 are eigen values of A and v_1, v_2 and v_3 are corresponding orthonormal eigen vectors of A and $P = [v_1\ v_2\ v_3]$ be an orthogonal matrix then $X^{T}P^{-1}APX = \lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2$ is called canonical form or principal axes form of the quadratic form.

Nature of the Quadratic form: A quadratic form Q is

1. Positive Definite if $Q(X) > 0\ \forall\ x \neq 0\ \Rightarrow \lambda_i > 0$ for all i.
2. Negative Definite if $Q(X) < 0\ \forall\ x \neq 0 \Rightarrow \lambda_i < 0$.
3. Positive Semidefinite if $Q(X) \geq 0\ \forall\ x \Rightarrow \lambda_i \geq 0$.
4. Negative Semidefinite if $Q(X) \leq 0\ \forall\ x \Rightarrow \lambda_i \leq 0$.
5. Indefinite if $Q(X)$ has both positive and negative values.

Index of a Quadratic form: The number of positive terms in its canonical form is called the index of a quadratic form.

Rank of the Quadratic form: The number of non-zero terms in its canonical form is called rank of the quadratic form.

Signature of the Quadratic form: Signature of the quadratic form is the difference of the positive and negative terms in the canonical form.

III. Determine the modal matrix which reduces the following quadratic forms to the canonical forms. Hence find nature, rank, signature and index of the quadratic forms:

1. $x^2 + 3y^2 + 3z^2 - 2yz$.
2. $3x^2 + 5y^2 + 3z^2 - 2xy + 2xz - 2yz$.
3. $x^2 + 5y^2 + z^2 + 2xy + 6xz + 2yz$.
4. $2xy + 2xz - 2yz$.
5. $3x^2 + 3y^2 + 3z^2 + 2xy + 2xz - 2yz$.
6. $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$

SINGULAR VALUE DECOMPOSITION

Let A be any matrix $m \times n$ with rank r then this matrix can be decomposed into product of three matrices given by $A_{m\times n} = U_{m\times m}\, \Sigma_{\mathrm{m\times n}}\, V_{n\times n}^{T}$ and it is called singular value decomposition of A. Here $\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$ is an $m \times n$ matrix for which the diagonal entries are r singular values of A, $\sigma_1 \geq \sigma_2 \geq \sigma_3 \ldots \sigma_r > 0$. U and V are orthogonal matrices or unitary matrices which represent the left singular and right singular vectors of A.

Working Rule:

1. For a given matrix A, compute $A^T A$ and obtain its eigenvalues and orthonormal eigenvectors.
2. Obtain singular values as $\sigma_i = \sqrt{\lambda_i}$ and arrange them in descending order to construct the diagonal matrix Σ.
3. Construct V from eigenvectors and compute U using $u_i = \frac{A\, v_i}{\sigma_i}$.
4. Hence, the singular value decomposition is given by $A = U\Sigma V^T$.

Note:

- If $A^T A$ has r non-zero eigenvalues then $A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \sigma_3 u_3 v_3^T + \cdots + \sigma_r u_r v_r^T$ which gives the r rank approximation of the data in the matrix A.
- The square root of eigen values of a symmetric matrix $A^T A$ or AA^T are called singular values of the matrix A.
- The eigen vectors of $A^T A$ corresponding to singular values of A are called singular vectors. The singular vectors of $A^T A$ are called right singular vectors and singular vectors of AA^T are called left singular vectors.

IV. Find the singular value decomposition of the following matrices

1. $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$.
2. $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$.
3. $A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$.
4. $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.
5. $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$
6. $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$.
7. Your USN is ABCD121212. Construct a 3×2 matrix by considering the last 6 digits of the USN. Perform the singular value decomposition for this matrix. Write your comments on the dimensionality reduction.

PRINCIPAL COMPONENT VARIABLES

Working Rule: To determine the principal components of a given dataset, follow the steps below:

Step 1: Given $M = [X\ Y]$, Compute the mean of each variable $\bar{X} = \frac{\Sigma X}{n}$, $\bar{Y} = \frac{\Sigma Y}{n}$ and form the centred matrix $\bar{M}$.

$$A = \bar{M} = [X - \bar{X} \quad Y - \bar{Y}].$$

Step 2: Determine the covariance matrix

$$C = \frac{1}{n-1} A^T A = \begin{bmatrix} cov(X,X) & cov(X,Y) \\ cov(Y,X) & cov(Y,Y) \end{bmatrix}.$$

Step 3: Compute all eigenvalues and corresponding eigenvectors of C.

Step 4: Arrange the eigenvalues in descending order and find the normalized eigen vectors. The normalized eigenvectors are called the principal components of the data.

Step 5: Obtain the first *r* principal components by transforming the data as

$$Y = e_i^T A = [PA_1 \;\; PA_2 \;\; \ldots \;\; PA_r].$$

The total variance = $Trace(C)$ then the percentage of variance of each principal component is given by $\frac{\lambda_i}{Trace(C)}$.

V. Find the principal components of the following data and hence reduce the data to the dimension one. Find percentage of variance of each principal component

1. $\begin{bmatrix} 19 & 22 & 6 & 3 & 2 & 20 \\ 12 & 6 & 9 & 15 & 13 & 5 \end{bmatrix}$

2. $\begin{bmatrix} 1 & 5 & 2 & 6 & 7 & 3 \\ 3 & 11 & 6 & 8 & 15 & 11 \end{bmatrix}$

3. $\begin{bmatrix} 4 & 11 \\ 8 & 4 \\ 13 & 5 \\ 7 & 14 \end{bmatrix}$

4. Find the principal components of the data, and list the new variable determined by the first principal component for the covariance matrix:

$$\begin{bmatrix} 2382.78 & 2611.84 & 2136.20 \\ 2611.84 & 3106.47 & 2553.90 \\ 2136.20 & 2553.90 & 2650.71 \end{bmatrix}$$

5. Given the data in table, reduce the dimension from 2 to1 using principal component analysis.

X	4	8	13	7
Y	11	4	5	14

6. Suppose three tests are administered to a random sample of college students. Let $\boldsymbol{X}_1 \ldots \boldsymbol{X}_N$ be observation vectors in R^3 that list the three scores of each student, and for $j = 1,2,3$ let x_j denote a student's score on the j^th exam. Suppose the covariance matrix of the data is

$$S = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 6 & 2 \\ 0 & 2 & 7 \end{bmatrix}$$

Let y be an "index" of student performance, with $y = c_1x_1 + c_2x_2 + c_3x_3$ and $c_1^2 + c_2^2 + c_3^2 = 1$. Choose c_1, c_2, c_3 so that the variance of y over the data set is as large as possible.